

UniConnect

CSE Departmental Hub

Software Requirements Specification

Version 2 | April 2026

Field	Details
Team	Team RAAS
Project	UniConnect (CSE Departmental Hub)
Repository	https://github.com/ashesh-2021911194-ctrl/uniConnect
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Authors

Name	ID / Email	Role
Sakib Hossain	sakib-2021311181@cs.du.ac.bd	Author
Ashesh Bar	ashesh-2021911194@cs.du.ac.bd	Author
Ruhul Amin	ruhul-2021811203@cs.du.ac.bd	Author
Abir Hasan	abir-2020415646@cs.du.ac.bd	Author



1. Document Control

1.1 Revision History

Version	Date	Author	Changes
0.1	07-Mar-2026	Development Team	Initial draft — SRS v1
1.0	08-Mar-2026	Development Team	Added CR upload types; new FRs, user stories, use cases, test cases
2.0	Apr-2026	Development Team	Major revision: replaced mandatory FB-Auth with optional social verification; added External Buyer FRs; added auth flows (password reset, logout, profile, account deletion, notifications); added CR role assignment/revocation; added Marketplace edit/delete; added Housing edit/delete; added Admin analytics; fixed roommate matching FR-test mismatch; added Housing use case UC-06 and Buyer-side Marketplace use case UC-07; clarified SOS cancellation, nearest/active definitions, and geocoding; resolved all v1 open issues



2. Problem Definition

2.1 Problem Statement

UniConnect addresses the fragmented digital infrastructure of the Computer Science & Engineering (CSE) department. Currently, academic resources are scattered across informal Google Drive links, Facebook groups, and physical notice boards — creating inefficiency, missed communications, and safety gaps.

The platform consolidates academic resources, peer-to-peer commerce, off-campus housing, and emergency safety protocols into a single verified, secure, and searchable application — acting as the 'Departmental Brain' for the CSE department.

2.2 Target Users

User Group	Description	Main Needs
CSE Students	Undergraduate/postgraduate students of the CSE department	Access resources, buy/sell items, find housing, trigger SOS
Class Representatives (CR)	Student representatives per batch; upload and manage all departmental content types	Upload lab tasks, slides, books, question papers, project showcases, course materials
Administrators	Department faculty/staff	Moderate content, manage user verification, view analytics

2.3 Goals and Non-Goals

Goals:

- Centralize academic resources (slides, question papers, books, lab sheets) in a searchable repository.
- Allow CRs to upload all resource types: lecture slides, reference books (PDF), previous year mid-term and final exam papers.
- Provide a secure peer-to-peer marketplace for CSE-specific items (laptops, Arduino kits, textbooks).
- Enable off-campus housing discovery with roommate matching based on class schedule overlap (shared courses/labs), with a CSE-Only visibility filter.
- Implement a real-time safety system (Panic Button + GPS SOS broadcast) for student safety, including SOS cancellation within a grace period.
- Enforce identity verification via university email. Social account linking (Facebook or Google) is optional and used only as an additional identity signal.
- Support complete account lifecycle: registration, login, password reset, profile management, logout, and account deletion.
- Assign and revoke the CR role through an Admin-managed workflow.
- Provide Admin analytics: resource uploads per batch, marketplace activity, SOS events, and user registration trends.

Non-Goals:

- UniConnect will NOT extend beyond the CSE department to other faculties.
- UniConnect will NOT replace the official university LMS (Learning Management System).
- UniConnect will NOT provide academic grading or assignment submission features.
- Facebook account linking is NOT mandatory for registration or login in v2. It is an optional social verification step.



3. Scope and Context

3.1 In Scope

- Module A: Academic Resource & Lab Archive — batch-wise repository, question bank (mid-term & final), lecture slides, reference books (PDF), lab central, project showcase, bilingual search.
- Module B: CSE Marketplace (Dorm-to-Dorm) — item listings with categories (hardware, books, software licenses); seller contact gating; listing creation, editing, deletion; buyer-side browsing; real-time chat system between buyer and seller for inquiries and negotiation before contact sharing.
- Module C: To-Let & Accommodation Finder — off-campus housing listings (create, edit, delete); roommate

matching by shared course/lab schedule overlap; CSE-Only visibility toggle; geocoded address input with map pin; availability toggles.

- Module D: Safety & Verified Social — SOS panic button with GPS broadcast; SOS cancellation within a 5-second grace window; definition of 'active' (logged in, app in foreground, last ping ≤ 5 min); definition of 'nearest' (within 2 km radius); proctorial directory.
- Module E: User Account Management — registration, email verification, login, password reset via email OTP, profile edit, logout, account deletion with data retention policy.
- Module F: CR Role Management — Admin assigns/revokes CR role; CR role scoped to a specific batch; CRs may upload for their assigned batch only.
- Module G: Admin Dashboard & Analytics — moderation queue; resource upload stats per batch/course; marketplace listing counts; SOS event log; user registration trend chart.
- REST API backend with JWT authentication and WebSocket support for real-time alerts.
- Mobile-responsive React/Next.js frontend.

3.2 Out of Scope

- In-app payment processing or escrow services.
- Integration with other university departments or faculties.
- Native mobile apps (iOS/Android) — web-responsive only in v1.
- Automated course scheduling or timetable generation.
- AI-based recommendation engines for resources or housing.

3.3 Assumptions and Constraints

Technical Constraints:

- Backend: Supabase (PostgreSQL database with built-in RESTful APIs and serverless functions).
- File storage: Supabase Storage for images and PDFs.
- Real-time: Supabase Realtime (WebSocket-based) for live features such as SOS broadcast and chat.
- Authentication: Supabase Auth (JWT-based sessions, secure password handling with built-in hashing). Facebook/Google OAuth is optional for social verification only.
- Uploaded files must be PDF format; maximum file size 25 MB per upload.
- Geocoding: user enters an address string; the system calls a geocoding API (e.g., Google Maps Geocoding API) to convert it to lat/lng, which is stored and used to render a map pin.

Organizational Constraints:

- Only verified university-domain email addresses can register as Students or CRs.
- Marketplace and housing contact info visible to verified CSE members only.
- Content moderation is required before any resource (slides, books, question papers, lab sheets) goes live.
- Only users with the CR role may upload academic resources; the CR role is batch-scoped.

Timeline Constraints:

- SRS v2 submission: April 2026.
- MVP delivery expected within one academic semester.



4. Stakeholders and Roles

4.1 Stakeholders

Stakeholder	Interest	Priority
CSE Students	Access to resources, safe marketplace, housing, emergency tools	High
Class Representatives	Upload and manage all departmental content types for their batch	High
Department Administration	Content governance, student safety oversight, usage analytics	High
University IT Dept.	Infrastructure, data security compliance	Medium
Parents / Guardians	Student safety via SOS alerts	Medium

4.2 User Roles and Permissions

Role	Description	Permissions
Student (Verified)	Registered CSE student with verified university email	Browse/download resources; create, edit, delete own marketplace listings; create, edit, delete own housing posts; trigger SOS; manage own profile; reset password; delete account
Class Representative (CR)	Designated batch representative, assigned by Admin. Scoped to one batch.	All Student permissions + upload, edit, delete lab sheets, lecture slides, reference books, mid-term papers, final papers, and project showcases for their assigned batch only
Admin	Department faculty or appointed moderator	All CR permissions + moderate/delete any content; manage user verification; assign/revoke CR role (scoped to batch); view analytics dashboard; manage data retention



5. Functional Requirements

Requirement ID format: FR-UC-###

5.1 Authentication & Account Management

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-001	User Registration	System shall allow registration using university-domain email only for Student/CR accounts.	M	Non-university email rejected with error message.
FR-UC-001C	Email Verification	System shall send a verification email with a 6-digit OTP on registration. Account is inactive until OTP is confirmed. OTP expires in 15 minutes.	M	Unverified account cannot log in; OTP confirmed within 15 min activates account.
FR-UC-002	Optional Social Verification	System shall allow (but NOT require) linking of a Facebook or Google account as an optional identity trust signal. Social linking is available in Profile Settings post-registration.	S	User can register and log in without social account linking. Linked social profile displays a verified badge on listings.
FR-UC-050	Login	System shall authenticate users via email + password. On success, issue a JWT access token (15 min TTL) and a refresh token (7 days TTL). Lock account for 15 minutes after 5 consecutive failed login attempts.	M	Valid credentials return JWT; 5 failed attempts trigger lockout with email notification.
FR-UC-051	Password Reset	System shall provide a password reset flow: user enters registered email; system sends a reset link (valid 1 hour); user sets new password; all active sessions are invalidated.	M	Reset link sent within 30 s; link expires after 1 hour; password updated; prior sessions invalidated.
FR-UC-052	Logout	System shall allow a user to log out, invalidating their current JWT refresh token server-side.	M	Post-logout, refresh token rejected; re-authentication required.
FR-UC-053	Profile Management	System shall allow verified users to view and edit their profile: display name, profile photo, bio, optional social account links, notification preferences.	M	Edits saved and reflected on the user's public listing cards within 5 seconds.
FR-UC-054	Account Deletion	System shall allow a user to request account deletion. Personal data (name, email, photo) is anonymised within 30 days. Content (listings, resources) is marked orphaned and retained for 90 days, then purged,	M	Deletion request triggers confirmation email; user cannot log in after confirmation; PII anonymised within 30 days.

		unless an Admin chooses to reassign it.		
FR-UC-055	Notifications	System shall send in-app and email notifications for: (a) upload approved/rejected by Admin, (b) marketplace item marked Sold, (c) new SOS alert received, (d) CR role assigned/revoked, (e) account lockout triggered. Users can opt out of email notifications per category in Profile Settings.	M	Notifications appear in the in-app bell icon within 5 s of event; email sent within 60 s; opt-out respected on next event.

5.2 CR Role Management

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-060	CR Role Assignment	System shall allow an Admin to assign the CR role to a verified Student, scoped to a specific batch. A student may hold the CR role for at most one batch at a time.	M	Assigned CR can upload resources for their batch; cannot upload for other batches; CR receives notification.
FR-UC-061	CR Role Revocation	System shall allow an Admin to revoke the CR role from a user. The user reverts to Student role. Resources previously uploaded by the user remain in the repository.	M	Revoked user loses upload access immediately; upload button hidden; notification sent; prior uploads intact.

5.3 Academic Resource & Archive

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-003	Batch Repository	System shall organize resources by Batch > Semester > Course hierarchy.	M	Navigation drill-down returns correct files.
FR-UC-004	Question Bank — Mid-Term	System shall provide a searchable database of previous year mid-term exam papers with tag filtering by course code and year.	M	Searching a course code returns only mid-term papers for that course; year filter works correctly.
FR-UC-004B	Question Bank — Final	System shall provide a searchable database of previous year final exam papers with tag filtering by course code and year.	M	Searching a course code returns only final papers; year and exam-type filters work correctly.
FR-UC-005	Lab Central	System shall store and serve lab task sheets per course; only CRs (for their batch) may upload.	M	CR uploads PDF; students can download after admin approval.
FR-UC-005B	Lecture Slides Upload	System shall allow CRs to upload lecture slide decks (PDF) tagged by Batch, Semester, Course, and Lecture number.	M	Uploaded slide PDF appears under correct course within 1 minute after admin approval; lecture number tag is searchable.
FR-UC-005C	Reference Books Upload	System shall allow CRs to upload reference book PDFs tagged by course code, author, and edition. CR must confirm they hold the right to share the material before upload.	M	Uploaded book PDF appears under correct course; author/edition metadata displayed; copyright consent checkbox recorded.
FR-UC-005D	Mid-Term Paper Upload	System shall allow CRs to upload previous year mid-term question papers tagged with course code, year, and exam type = Mid-Term.	M	Uploaded paper appears in the Question Bank under Mid-Term filter for the correct course and year.
FR-UC-005E	Final Exam Paper Upload	System shall allow CRs to upload previous year final exam question papers tagged with course code, year, and exam type = Final.	M	Uploaded paper appears in the Question Bank under Final filter for the correct course and year.
FR-UC-006	Project Showcase	System shall display a Wall of Fame linking student projects to GitHub repositories.	S	Valid GitHub URL required on submission.
FR-UC-007	Bilingual Search	System shall support searching course materials in both English and native script (Bengali).	S	Native-language query returns same results as English.
FR-UC-016	Upload Type Tagging	System shall enforce mandatory metadata tagging on every upload: resource type (Slide/Book/MidTerm/Final/Lab), course code, batch, semester. Year is additionally required for Mid-Term and Final paper uploads.	M	Upload rejected with validation error if any required tag is missing.
FR-UC-017	Duplicate Upload Detection	System shall warn CR when an identical filename + course code + year combination already exists in the repository.	S	Duplicate warning shown; CR can choose to overwrite or cancel.

5.4 Marketplace

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-008	Create Listing	System shall allow verified users to post a marketplace item with name, price, condition, category, and up to 5 images.	M	Non-logged-in users cannot view seller contact info.
FR-UC-008B	Edit Listing	System shall allow the listing owner (or Admin) to edit any field on an existing marketplace listing.	M	Edited listing reflects updates within 5 seconds; edit history logged server-side.
FR-UC-008C	Delete Listing	System shall allow the listing owner (or Admin) to delete a marketplace listing. Deleted listings are soft-deleted and hidden from public view; retained for 30 days then purged.	M	Deleted listing no longer appears in search; confirmation dialog shown before deletion.
FR-UC-009	Marketplace Categories	System shall filter marketplace items by Hardware, Software Licenses, and Academic Books.	M	Category filter returns only items of that type.
FR-UC-009B	Buyer Browse & Contact	Verified CSE users may browse listings, view item details, and directly initiate in-app chat with the seller for inquiries and negotiation.	M	Users see a "Chat with Seller" option on listings; the seller receives a notification, and both parties can communicate through the built-in messaging system.

5.5 Housing & To-Let

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-010	Create Housing Listing	System shall allow users to post off-campus housing listings with rent, type, and a text address. The system geocodes the address via a geocoding API to obtain lat/lng for the map pin.	M	Listing appears on map with correct coordinates; geocoding failure shows error and allows manual lat/lng entry.
FR-UC-010B	Edit Housing Listing	System shall allow the listing owner (or Admin) to edit any field on an existing housing listing.	M	Edits reflected within 5 seconds; map pin updates if address changed.
FR-UC-010C	Delete Housing Listing	System shall allow the listing owner (or Admin) to delete a housing listing. Soft-deleted; hidden from public view; purged after 30 days.	M	Deleted listing removed from map and list; confirmation dialog shown.
FR-UC-011	Roommate Matching	System shall allow users to indicate shared-schedule roommate preference. The system filters roommate requests by course/lab schedule overlap: two users are considered schedule-compatible if they share at least one course or lab section in the current semester. A separate CSE-Only toggle filters to CSE-department users only.	S	Schedule-compatible filter returns only users sharing ≥ 1 course/lab section; CSE-Only toggle hides non-CSE listings independently.
FR-UC-012	Availability Toggle	System shall provide real-time Available/Occupied status for housing listings.	M	Toggle update reflected within 5 seconds.

5.6 Safety

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-013	SOS Panic Button	System shall broadcast user GPS coordinates to nearest active CSE students on 3-second hold. 'Active' = logged in, app in foreground within last 5 minutes. 'Nearest' = within a 2 km radius. If no users are within 2 km, the system broadcasts to the 5 most-recently-active CSE users regardless of distance.	M	Alert reaches active users within 2 seconds of trigger; nearest definition verified in testing.
FR-UC-013B	SOS Cancellation	System shall allow the triggering user to cancel an SOS alert within a 5-second grace window after the 3-second hold. If cancelled within 5 seconds, the alert is suppressed and not broadcast.	M	Cancel within 5 s suppresses broadcast; cancel after 5 s has no effect (alert already sent); event log records both triggered and cancelled status.
FR-UC-014	Proctorial Directory	System shall provide a one-tap call list for Campus Security, Medical, and Dept. Head.	M	Directory loads within 2 seconds; calls initiate correctly.

5.7 Content Moderation & Admin

Req ID	Feature	Requirement Statement	Pri	Acceptance Criteria
FR-UC-015	Content Moderation	System shall allow Admins to review, approve, or delete all uploaded content including slides, books, and question papers.	M	Unapproved content not visible to students.
FR-UC-070	Admin Analytics Dashboard	System shall provide an Admin analytics dashboard showing: (a) resource uploads per batch/course/type over time, (b) marketplace listing count and sold rate, (c) SOS event log with timestamp, coordinates, and resolution status, (d) user registration trend (daily new accounts over the last 90 days).	M	Dashboard loads within 3 seconds; each chart shows correct aggregated data; SOS log exportable as CSV.



6. User Stories and Use Cases

6.1 User Stories

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-UC-030	New User	to reset my forgotten password via email OTP	I can regain access without contacting Admin	Reset link sent within 30 s; password updated; prior sessions invalidated
US-UC-031	Verified User	to update my display name, photo, and notification preferences	My profile reflects who I am and I control what emails I receive	Profile edits saved and reflected in listings within 5 seconds
US-UC-032	Verified User	to permanently delete my account and have my personal data anonymised	I can leave the platform with confidence my data is removed	PII anonymised within 30 days; login no longer possible
US-UC-033	Verified User	to browse marketplace listings, and submit a contact request to a seller	I can buy departmental items	marketplace browseable; seller notified of contact request; seller can accept/decline
US-UC-034	Admin	to assign the CR role to a verified student scoped to a specific batch	Only designated representatives can upload content for that batch	Assigned CR can upload for their batch only; other batches blocked
US-UC-035	Admin	to revoke the CR role from a student	Students who leave the role lose upload privileges immediately	Upload access removed immediately; notification sent to revoked user
US-UC-001	CSE Student	to browse resources by batch and course	I can find relevant slides quickly	Resource tree navigable in <= 3 clicks to file
US-UC-002	CSE Student	to search for past mid-term exam papers by course tag and year	I can prepare for mid-terms efficiently	Tag + year filter returns only matching mid-term papers
US-UC-002B	CSE Student	to search for past final exam papers by course tag and year	I can prepare for finals efficiently	Tag + year filter returns only matching final exam papers
US-UC-003	CSE Student	to download lab sheets for my enrolled courses	I can complete labs without asking seniors	Download link active; file opens correctly
US-UC-003B	CSE Student	to download lecture slides for my courses	I can study without relying on informal sharing groups	Slide PDF downloads correctly; tagged with correct lecture number
US-UC-003C	CSE Student	to download reference book PDFs for my courses	I have access to course books without buying them	Book PDF downloads with author and edition shown
US-UC-004	CSE Student	to view student projects with GitHub links	I can learn from senior projects	Each project card shows title, description, and working GitHub link
US-UC-011	Class Rep (CR)	to upload lab task sheets for my batch	Students have structured access to lab materials	Uploaded file appears under correct batch/course within 1 minute after approval
US-UC-011B	Class Rep (CR)	to upload lecture slides tagged by course and lecture number	Students can access slides in an organised, searchable way	Slide appears under correct course; lecture number tag is searchable
US-UC-011C	Class Rep (CR)	to upload previous year mid-term question papers tagged with course code and year	Students can find past mid-term papers quickly	Paper appears in Question Bank under Mid-Term filter for correct course and year
US-UC-011D	Class Rep (CR)	to upload previous year final exam question papers tagged with course code and year	Students can find past final papers quickly	Paper appears in Question Bank under Final filter for correct course and year
US-UC-011E	Class Rep (CR)	to upload reference book PDFs with author	Students have one-stop access to course books	Book appears under correct course; metadata

		and edition metadata, and confirm I have the right to share		displayed; copyright checkbox recorded
US-UC-005	CSE Student	to list a used laptop for sale	I can sell to trusted department peers	Listing visible to verified users only; seller contact hidden from guests
US-UC-005B	CSE Student	to edit or delete my marketplace listing	I can keep my listing accurate or remove it when the item is gone	Edit reflected within 5 s; deletion hides listing with confirmation dialog
US-UC-006	CSE Student	to filter marketplace by hardware category	I find only relevant items	Category filter shows only hardware items
US-UC-007	CSE Student	to post a housing listing by typing my address and having the system place a map pin	Batchmates can find accommodation near campus	Map pin renders at correct geocoded coordinates; geocoding failure shown gracefully
US-UC-007B	CSE Student	to edit or delete my housing listing	I can update or remove my post when circumstances change	Edit reflected within 5 s; deletion confirmed and listing hidden
US-UC-008	CSE Student	to find a roommate who shares at least one of my courses or labs this semester	I avoid schedule conflicts with my roommate	Schedule-compatible filter returns only users sharing ≥ 1 course/lab section in current semester
US-UC-009	CSE Student	to trigger an SOS alert in an emergency and cancel it within 5 seconds if triggered by mistake	Nearby students and security are notified, but accidental triggers are preventable	Alert dispatched within 2 s; cancellation within 5-second grace window suppresses broadcast
US-UC-010	CSE Student	to access emergency contact numbers with one tap	I can reach security or medical staff fast	Directory loads within 2 seconds; tap initiates call
US-UC-013	Admin	to review and approve uploaded resources before publication	Only verified, quality content reaches students	Pending items listed in moderation queue; approve/reject works
US-UC-036	Admin	to view an analytics dashboard showing uploads, marketplace activity, SOS events, and registration trends	I can make informed decisions about platform governance	Dashboard loads in ≤ 3 s; data is accurate; SOS log exportable as CSV
US-UC-016	CSE Student	to search for course materials in Bengali script	I can find resources regardless of which language I search in	Native-language query returns same results as English equivalent



6.2 Critical Use Cases

UC-01: Student Accesses Academic Resource

Field	Detail
Preconditions	User is logged in with verified university email.
Main Flow	1. User navigates Home > Archive Tab. 2. Selects Batch > Semester > Course. 3. Selects resource type tab: Slides Books Question Papers (Mid-Term / Final) Lab Sheets. 4. Browses available PDFs. 5. Clicks Download to save file.
Alternate Flow	If no resources exist for the course, system shows 'No resources uploaded yet' placeholder.
Exception Flow	If file URL is broken, system shows error and logs the issue to admin queue.
Postconditions	File is downloaded; download count incremented in DB.

UC-02: Student Triggers SOS Panic Alert

Field	Detail
Preconditions	User is logged in, GPS permissions granted, active internet connection.
Main Flow	1. User holds Panic Button for 3 seconds from any screen. 2. A 5-second countdown overlay appears with a 'Cancel' button. 3. If not cancelled within 5 seconds, emergency overlay activates and the system broadcasts GPS coordinates via WebSocket to CSE users within 2 km who are active (last ping ≤ 5 min, app in foreground). 4. SOS notification delivered within 2 seconds of broadcast. 5. Alert logged with timestamp, coordinates, and status = Active.
Alternate Flow A	If GPS unavailable, system broadcasts last known location with a warning flag.
Alternate Flow B	If no active users are within 2 km, system broadcasts to the 5 most-recently-active CSE users regardless of distance.
Alternate Flow C	If user cancels within the 5-second grace window, the overlay closes, no broadcast is sent, and the event is logged as Cancelled.
Exception Flow	If WebSocket connection fails, system attempts HTTP fallback push notification.
Postconditions	Alert logged with timestamp, coordinates, and final status (Active or Cancelled); nearby users notified if Active.

UC-03: CR Uploads Academic Resource

Field	Detail
Preconditions	User is logged in with CR role scoped to a batch; file is PDF; file size ≤ 25 MB.
Main Flow	1. CR navigates to Upload section. 2. Selects resource type from dropdown: Lab Sheet

	Lecture Slide Reference Book Mid-Term Paper Final Paper. 3. System dynamically shows relevant metadata fields for the chosen type. 4. System validates all required tag fields are filled. 5. System checks for duplicates (same filename + course code + year). 6. For Reference Book: CR ticks copyright consent checkbox. 7. CR uploads PDF to cloud storage. 8. Admin receives moderation notification with resource type and tags.
Alternate Flow	If file exceeds 25 MB: upload rejected with size error. If duplicate detected: CR warned and can overwrite or cancel. If cloud storage fails: system queues upload and retries within 5 minutes; CR sees 'Upload queued' notification.
Exception Flow	If CR attempts to upload for a batch other than their assigned batch, the system rejects with 'Unauthorised batch' error.
Postconditions	File stored in cloud under correct resource-type category; pending admin approval; not visible to students until approved.

UC-04: Student Posts Marketplace Item

Field	Detail
Preconditions	User is logged in with verified university email.
Main Flow	1. User navigates Home > Market Tab > 'Post Item'. 2. Fills in item name, price, condition, category (ENUM), and uploads up to 5 images. 3. System validates price > 0 and category is valid. 4. Listing saved with status = Available; visible to all verified users. 5. Seller can receive in-app chat requests/messages from interested buyers via "Chat with Seller" option.
Alternate Flow	If image upload fails, user can submit listing without images and add later.
Exception Flow	If price field is 0 or negative, system shows: 'Price must be greater than 0'.
Postconditions	item appears in marketplace and is visible only to verified users; buyers can initiate private chat with the seller, and conversations are visible only to the participating buyer and seller.

UC-05: Admin Moderates Uploaded Content

Field	Detail
Preconditions	Admin is logged in; at least one resource or project is pending in moderation queue.
Main Flow	1. Admin navigates to Admin Dashboard > Moderation Queue. 2. Reviews pending item (file preview, uploader ID, resource type, course tag, year). 3. Clicks 'Approve' to publish or 'Reject' with reason. 4. System moves approved item to student-facing archive; rejected item flagged and uploader notified.
Alternate Flow	If queue is empty, dashboard shows 'No items pending review'.
Exception Flow	If Admin accidentally approves wrong item, Admin can use 'Remove' action post-approval to hide it.
Postconditions	Approved content visible to students under correct type tab; rejected item removed from queue; uploader email notification sent.

UC-06: Student Creates and Manages Housing Listing

Field	Detail
Preconditions	User is logged in with verified university email.
Main Flow	1. User navigates Home > To-Let Tab > 'Post Listing'. 2. Fills in listing type (Sublet/Flat/Roommate), rent, number of rooms, and a text address. 3. System calls geocoding API to convert address to lat/lng. 4. Map preview shows the pin at the resolved location. 5. User confirms coordinates and submits. 6. Listing saved with status = Available, visible to CSE users (and optionally restricted via is_cse_only toggle).
Alternate Flow A	If geocoding fails, system shows error and offers a manual lat/lng input fallback.
Alternate Flow B	Owner returns to listing and clicks 'Edit' to update any field; map pin updates if address changes.
Alternate Flow C	Owner clicks 'Delete'; confirmation dialog shown; listing soft-deleted and hidden from map/list.
Postconditions	Listing visible on map at correct coordinates; owner can edit or delete at any time.

UC-07: Buyer Browses Marketplace and Contacts Seller

Field	Detail
Preconditions	User is logged in with a verified university email.
Main Flow (CSE User)	1. User navigates Home > Market Tab. 2. Applies category filter (e.g., Hardware). 3. Clicks item card to view details (name, price, condition, images, seller display name). 4. Clicks "Chat with Seller". 5. System creates (or opens existing) private chat conversation between buyer and seller. 6. Seller receives a notification of the new message. 7. Buyer and seller communicate through the in-app messaging system.
Exception Flow	If seller does not respond within 7 days, request expires automatically and buyer is notified. If message delivery fails (e.g., network issue), system shows: "Message not sent. Please try again."
Postconditions	A private one-to-one chat is established between buyer and seller; messages are visible only to the participants and not accessible to other users.



7. Data Requirements

7.1 Core Entities

Entity	Description	Key Fields	Constraints
USER	Registered platform member (CSE student, CR, Admin, or external buyer)	user_id (PK), email, is_verified, is_external, role (ENUM), social_uid (nullable), batch_id (FK, nullable)	Email unique; role ENUM: Student, CR, Admin, External; social_uid nullable (optional linking)
BATCH	Academic batch grouping	batch_id (PK), batch_year, department	References multiple USERS and PROJECTs
CR_ASSIGNMENT	Tracks CR role scoped to a batch	assignment_id (PK), user_id (FK), batch_id (FK), assigned_by (FK to Admin), assigned_at, revoked_at (nullable)	One active assignment per user; revoked_at null = currently active
RESOURCE	Uploaded academic file (any type)	resource_id (PK), file_url, resource_type (ENUM), course_code, uploader_id (FK), year (nullable), metadata (JSONB), copyright_consent (BOOLEAN)	resource_type ENUM: Lab_Sheet, Slide, Book, MidTerm_Paper, Final_Paper; year required for MidTerm/Final; copyright_consent required for Book
PROJECT	Student project showcase entry	project_id (PK), title, github_link, batch_id (FK)	github_link must match GitHub URL pattern
MARKET_ITEM	Marketplace listing item	item_id (PK), item_name, price, condition, category (ENUM), status (ENUM), seller_id (FK), deleted_at (nullable)	price > 0; status ENUM: Available, Sold; deleted_at null = active; soft-delete retained 30 days
CONTACT_REQUEST	Gated contact request from external buyer to seller	request_id (PK), item_id (FK), buyer_id (FK), seller_id (FK), status (ENUM), expires_at	status ENUM: Pending, Accepted, Declined, Expired; expires_at = created_at + 7 days
HOUSING_POST	Off-campus housing or roommate post	post_id (PK), type (ENUM), rent, address_text, location_coords (JSONB), is_cse_only, status, schedule_id (FK, nullable), deleted_at (nullable)	type ENUM: Sublet/Flat/Roommate; location_coords = {lat, lng}; deleted_at null = active
SCHEDULE	User's current-semester course/lab enrolment for roommate matching	schedule_id (PK), user_id (FK), semester, courses (JSONB array of course codes and section IDs)	courses array must contain at least 1 entry; used for overlap computation
SOS_EVENT	SOS alert log entry	event_id (PK), user_id (FK), triggered_at, coords (JSONB), status (ENUM), cancelled_at (nullable), resolution_note (nullable)	status ENUM: Active, Cancelled, Resolved; cancelled_at set if cancelled within grace window
NOTIFICATION	In-app notification record	notif_id (PK), user_id (FK), category (ENUM), message, is_read, created_at	category ENUM: UploadApproved, UploadRejected, SOSAlert, CRAssigned, CRRevoked, AccountLock, ContactRequest

7.2 RESOURCE Entity — Unified Type Model

Resource Type	Required Metadata Fields	Stored Under
Lab Sheet	batch, semester, course_code	Archive > Course > Lab Sheets
Lecture Slide	batch, semester, course_code, lecture_no	Archive > Course > Slides
Reference Book	course_code, author, edition, copyright_consent=true	Archive > Course > Books
Mid-Term Paper	course_code, year, exam_type=Mid-Term	Question Bank > Mid-Term
Final Paper	course_code, year, exam_type=Final	Question Bank > Final



8. Non-Functional Requirements

NFR ID format: NFR-UC-###

NFR ID	Category	Requirement	Target / Metric
NFR-UC-001	Performance	API response time for resource queries	<= 500 ms for 95% of requests under normal load
NFR-UC-002	Performance	SOS alert delivery latency	Active users notified within 2 seconds of 3-second hold trigger

NFR-UC-003	Security	All API endpoints require JWT Bearer token authentication	401 Unauthorized returned for missing/invalid tokens
NFR-UC-004	Security	Seller contact info hidden from unverified users	0% exposure rate to unverified/guest users in testing
NFR-UC-005	Security	User passwords stored as hashed values	bcrypt with minimum 12 salt rounds
NFR-UC-006	Usability	New user onboarding completion rate	>= 80% of users complete registration without external help
NFR-UC-007	Availability	Platform uptime during academic semester	>= 99% uptime during peak hours (8 AM–10 PM)
NFR-UC-008	Scalability	System handles concurrent active users	Minimum 500 concurrent users without performance degradation
NFR-UC-009	Data Integrity	File uploads validated before storage; resource_type tag validated server-side	Rejected files logged; no corrupted files stored in S3/Cloudinary
NFR-UC-010	Compliance	User data stored in compliance with institutional data policy	PII anonymised within 30 days of account deletion request; no PII shared with third parties beyond optional OAuth flow
NFR-UC-011	Storage	Per-user upload quota enforced to prevent cost overrun	Max 25 MB per file; max 500 MB total uploads per CR per semester
NFR-UC-012	Accessibility	WCAG AA contrast and keyboard navigation	Min 4.5:1 contrast ratio; all core flows keyboard-navigable; min font size 14px
NFR-UC-013	Responsiveness	All screens adapt for mobile, tablet, and desktop	320px+ (mobile), 768px+ (tablet), 1024px+ (desktop)



9. Risks and Open Questions

9.1 Risk Register

Risk ID	Description	Prob.	Impact	Mitigation
RISK-UC-001	Optional Facebook/Google OAuth fails or API terms change	Medium	Low	Social linking is optional; primary auth uses email + password only — OAuth failure does not break registration or login
RISK-UC-002	GPS spoofing on SOS panic button triggers false emergency broadcasts	Low	High	Rate-limit SOS triggers (max 3 per hour per user); 5-second cancellation window; log all SOS events with coordinates; Admin can mark event as False Alarm
RISK-UC-003	Low student adoption due to change from existing Facebook groups	High	High	Launch with CR-driven content seeding; onboarding incentives (resource access gating); in-app tutorial for first-time users
RISK-UC-004	Cloud storage cost overrun from unrestricted file uploads	Medium	Medium	Enforce 25 MB per-file limit; 500 MB per-CR per-semester quota (NFR-UC-011); Admin alert when total storage exceeds 80% of budget
RISK-UC-005	Non-CSE users bypass CSE-Only filter for housing/marketplace	Low	Medium	Server-side validation of is_verified and is_external on every protected endpoint; no client-side-only enforcement
RISK-UC-006	WebSocket server overload during mass SOS alert scenario	Low	High	Horizontal scaling with load balancer; Redis message queue for broadcast routing; HTTP fallback push notification
RISK-UC-007	CRs upload incorrect resource type (e.g., final paper tagged as mid-term)	Medium	Medium	Mandatory type tagging (FR-UC-016); Admin reviews type + tags during moderation; uploader can be notified of tag errors pre-approval
RISK-UC-008	Copyright concerns over uploaded book PDFs	Medium	High	Mandatory copyright consent checkbox (FR-UC-005C); Admin reviews for copyright issues in moderation queue; DMCA takedown process documented in ToS
RISK-UC-009	Geocoding API unavailability leaves housing posts without coordinates	Low	Medium	UC-06 Alternate Flow A: manual lat/lng fallback input; cached last-known result used if API is temporarily down; fallback to secondary geocoding provider
RISK-UC-010	Account lockout mechanism abused to lock out legitimate users via repeated failed attempts	Low	Medium	Lockout is per-IP as well as per-account; email notification sent on lockout; Admin can manually unlock

9.2 Resolved Questions (from SRS v1)

Question	Resolution
Is Facebook Auth mandatory?	No. Social account linking (Facebook or Google) is optional in v2 and used only as an identity trust signal. Primary auth is email + password.
Should the Panic Button require a confirmation step?	Yes — a 5-second cancellation grace window is shown after the 3-second hold, balancing speed with accidental-trigger prevention (FR-UC-013B).
How are map coordinates input for housing?	User enters a text address; the system geocodes it via a third-party geocoding API. Manual lat/ing fallback is available if geocoding fails (FR-UC-010, UC-06).
What does 'nearest active' mean for SOS?	'Active' = logged in, app in foreground, last heartbeat ping within 5 minutes. 'Nearest' = within 2 km. If nobody within 2 km, broadcast to 5 most-recently-active users (FR-UC-013).
Should a single CR upload for any batch?	No. CR role is batch-scoped. A CR may only upload for their assigned batch. Admin assigns the role per batch (FR-UC-060, FR-UC-061).
Should book PDFs require ISBN/copyright check?	Mandatory copyright consent checkbox added to book upload form. No ISBN required but recommended guidance shown. Admin reviews during moderation (FR-UC-005C, RISK-UC-008).
Is bilingual search required at MVP?	Designated as Should-Have (priority S) — included in scope but deferred to end of semester if time-constrained (FR-UC-007).
What is the data retention policy for deleted content?	Soft-deleted marketplace and housing content retained 30 days then purged. Account PII anonymised within 30 days of deletion request. Orphaned resources from deleted accounts retained 90 days then purged unless reassigned by Admin (FR-UC-054).

9.3 Remaining Open Questions

- Should the marketplace support in-app messaging, or rely entirely on FB Messenger redirect / email contact?
- Will the university provide an official API for real-time student ID or enrolment verification, or will manual email-domain verification remain the primary mechanism?
- Should the Admin analytics dashboard data be exportable in formats beyond CSV (e.g., PDF report)?



10. Testability and Traceability

10.1 Requirement Traceability Matrix

Req ID	Story / Use Case	Planned Test ID
FR-UC-001	US-UC-030 — University email enforcement	TC-UC-001
FR-UC-001C	US-UC-030 — Email OTP verification	TC-UC-001C
FR-UC-002	US-UC-031 — Optional social linking	TC-UC-002
FR-UC-050	Login + account lockout	TC-UC-050, TC-UC-050B
FR-UC-051	US-UC-030 — Password reset flow	TC-UC-051
FR-UC-052	Logout invalidates refresh token	TC-UC-052
FR-UC-053	US-UC-031 — Profile management	TC-UC-053
FR-UC-054	US-UC-032 — Account deletion	TC-UC-054
FR-UC-055	Notification delivery per category	TC-UC-055
FR-UC-060	US-UC-034 — CR role assignment	TC-UC-060
FR-UC-061	US-UC-035 — CR role revocation	TC-UC-061
FR-UC-003	US-UC-001 — Batch/Semester/Course navigation	TC-UC-003
FR-UC-004 / 004B	US-UC-002 / 002B — Question bank filters	TC-UC-004, TC-UC-004B
FR-UC-005	US-UC-011 — CR uploads lab sheet	TC-UC-005
FR-UC-005B	US-UC-011B — CR uploads lecture slides	TC-UC-005B
FR-UC-005C	US-UC-011E — CR uploads reference book with consent	TC-UC-005C
FR-UC-005D	US-UC-011C — CR uploads mid-term paper	TC-UC-005D
FR-UC-005E	US-UC-011D — CR uploads final paper	TC-UC-005E
FR-UC-008B/C	US-UC-005B — Edit/delete marketplace listing	TC-UC-008B, TC-UC-008C
FR-UC-009B	US-UC-033, UC-07 — buyer contact request flow	TC-UC-009B
FR-UC-010 / 010B / 010C	US-UC-007 / 007B, UC-06 — Housing CRUD + geocoding	TC-UC-010, TC-UC-010B, TC-UC-010C
FR-UC-011	US-UC-008 — Schedule-based roommate matching	TC-UC-011
FR-UC-013B	US-UC-009 — SOS cancellation grace window	TC-UC-013B
FR-UC-070	US-UC-036 — Admin analytics dashboard	TC-UC-070



10.2 Manual Test Plan

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-UC-001	Non-university email rejected at registration	User at registration page	Enter non-@university.edu email; submit	Error: 'Must use university email'
TC-UC-001C	Unverified account cannot log in; OTP expires after 15 min	User registered but OTP not entered	Attempt login; then wait 16 min and enter OTP	Login blocked pre-verification; OTP expired error after 15 min
TC-UC-002	Social account linking is optional; login succeeds without it	User registered without social link	Complete registration; skip social linking; attempt login	Login succeeds; no social badge on profile
TC-UC-050	5 failed login attempts trigger 15-min lockout	Valid account exists	Enter wrong password 5 times	Account locked; email notification sent; correct password fails during lockout
TC-UC-051	Password reset flow: link sent, expires in 1 hour, prior sessions invalidated	User with valid email	Request reset; click link within 60 min; set new password	Password updated; old JWT refresh token rejected; user must log in again
TC-UC-052	Logout invalidates refresh token	User logged in with active refresh token	Click Logout; attempt to use refresh token to get new access token	Refresh token rejected with 401; user redirected to login
TC-UC-053	Profile edit reflected on listing cards within 5 seconds	User with active marketplace listing	Edit display name in profile; save; view marketplace listing	New name appears on listing card within 5 seconds
TC-UC-054	Account deletion anonymises PII and blocks login	User with active account	Request deletion; confirm via email; attempt login after 30 days	Login blocked immediately after confirmation; PII anonymised within 30 days
TC-UC-055	Upload-approved notification appears in-app within 5 seconds	CR with pending upload; Admin approves it	Admin clicks Approve; CR checks notification bell	Notification appears within 5 seconds; email sent within 60 seconds
TC-UC-060	CR role assignment scoped to batch; other batches blocked	Admin logged in; target user is a verified student	Assign CR role for Batch 2023; CR attempts to upload for Batch 2022	Upload for Batch 2023 succeeds; upload for Batch 2022 rejected with 'Unauthorised batch' error
TC-UC-061	CR role revocation removes upload access immediately	CR with active role	Admin revokes CR role; CR attempts to upload immediately after	Upload button hidden; API returns 403; notification sent to revoked user
TC-UC-003	Resource navigation: Batch > Semester > Course > Type Tab	User logged in; resources uploaded	Navigate Archive > Batch > Semester > Course > click Slides tab	Only slide PDFs listed under selected course
TC-UC-004	Mid-term question bank year + tag filter	Mid-term papers uploaded with year tags	Select course='CSE-301'; year='2023'; type='Mid-Term'	Only CSE-301 mid-term papers from 2023 returned
TC-UC-005C	CR uploads reference book with copyright consent checkbox	CR logged in with CR role	Select type 'Reference Book'; fill course, author, edition; tick consent checkbox; upload	Book appears under course Books tab; consent recorded; upload rejected if consent not ticked
TC-UC-008B	Seller edits listing; update reflected within 5 seconds	Seller with active listing	Edit item price from 1500 to 1200; save	Updated price visible within 5 seconds
TC-UC-008C	Seller deletes listing; confirmation dialog shown; listing hidden	Seller with active listing	Click Delete on listing; confirm in dialog	Listing disappears from marketplace immediately; still in DB with deleted_at set
TC-UC-009B	Buyer contact request flow: seller receives notification and can accept/decline	Buyer account; active listing	Buyer clicks 'Request Contact'; seller checks notifications; clicks 'Share Contact'	Seller notified within 5 s; buyer receives seller contact info via email after acceptance
TC-UC-010	Housing listing geocodes address to map pin correctly	User logged in	Enter address 'ECE Building, Curzon Hall, Dhaka'; submit	Map pin rendered at correct lat/lng; coordinates stored in DB
TC-UC-010B	Housing listing edit updates map pin when address changes	Active housing listing with address A	Edit address to address B; save	Map pin moves to geocoded location of address B within 5 seconds
TC-UC-011	Roommate matching: schedule-compatible filter returns users sharing >= 1 course section	User A enrolled in CSE-301-S1 and CSE-401; User B enrolled in CSE-301-S1 and CSE-502; User C enrolled in CSE-501 only	User A applies 'Schedule Compatible' filter	User B appears in results (shares CSE-301-S1); User C does not appear
TC-UC-013	SOS alert reaches active nearby users within 2 seconds	User A logged in with GPS; User B active within 2 km (last ping < 5 min,	User A holds Panic Button for 3 seconds; does not cancel within 5 s	User B receives alert notification within 2 seconds; SOS_EVENT logged as Active

		app in foreground)		
TC-UC-013B	SOS cancellation within 5-second grace window suppresses broadcast	User A logged in with GPS	User A holds Panic Button for 3 seconds; clicks Cancel within 4 seconds of countdown start	No alert broadcast sent; SOS_EVENT logged as Cancelled
TC-UC-015	Admin moderation: approved resource becomes visible to students	CR uploaded resource; Admin on moderation queue	Admin clicks Approve on pending resource	Resource appears in student-facing archive under correct type tab immediately
TC-UC-070	Admin analytics dashboard loads in <= 3 seconds with correct data	Admin logged in; multiple uploads, listings, and SOS events exist	Navigate to Admin Dashboard > Analytics	Dashboard loads in <= 3 s; upload chart, listing count, SOS log, and registration trend all display; SOS log exports as CSV
TC-UC-017	Upload rejected if required tag is missing	CR on upload form	Select 'Mid-Term Paper'; leave year blank; submit	Error: 'Year is required for Mid-Term Paper uploads'
TC-UC-018	Duplicate upload warning shown	Identical file already uploaded	CR uploads file with same name + course + year as existing entry	Warning shown: 'This resource may already exist. Overwrite or Cancel?'



11. SRS v2 Submission Checklist

#	Item	Status
1	Problem statement and scope are clear	Done — Section 2, 3
2	Minimum 12 user stories with acceptance criteria	Done — Section 6.1 (30 stories included)
3	Functional requirements table is complete	Done — Section 5 (30+ requirements across 7 modules)
4	Data model and data dictionary are attached	Done — Section 7 (11 entities, unified RESOURCE type model)
5	Minimum 5 wireframes or flow screens included (original diagrams retained)	Done — Original document diagrams; see parent SRS v1 Section 8 appendix
6	NFR section is completed	Done — Section 8 (13 NFRs)
7	Risks and open questions are documented	Done — Section 9 (10 risks; 8 resolved questions; 4 remaining open questions)
8	Initial manual test plan (15+ cases) is included	Done — Section 10.2 (30 test cases covering all modules)
9	All v1 contradictions and gaps addressed	Done — FB-Auth made optional; auth flows complete; CR role management added; CRUD for listings; Admin analytics; SOS cancellation; geocoding; schedule-based matching; Housing UC-06 and Buyer UC-07 added

End of Document — UniConnect SRS v2